Draw only the forces on your chosen object

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Free-body diagram / ES: diagrama de cuerpo libre / PT: diagrama de corpo livre

Only the forces acting on one chosen object.

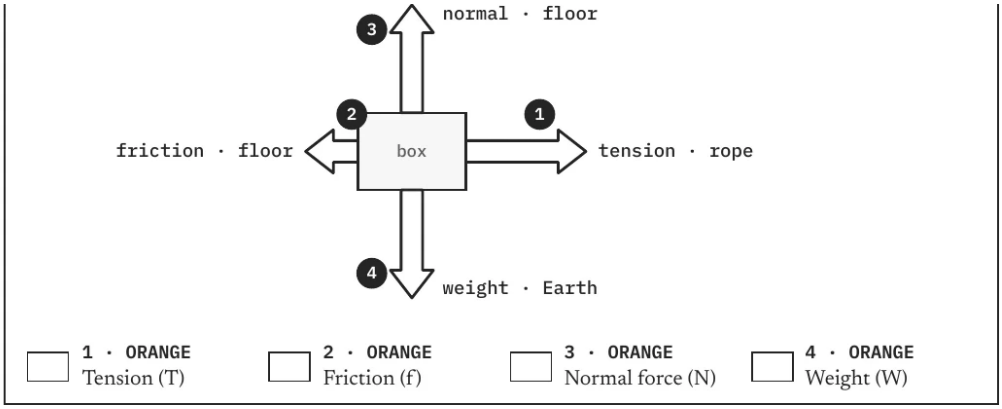
Also on your note page: Selected object, Interacting object, Force source.

B READ THE RELATIONSHIP

A free-body diagram shows the external forces on one chosen object.

Choose the receiver before drawing forces

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 20 N lamp hangs at rest from one vertical rope. Draw and label its forces.

Given weight = 20 N at rest one rope

Rule sources: Earth, rope

Work weight 20 N down tension up, balances it

Answer Tension 20 N up equal-length arrows

E SAVE FOR THE FINAL MEETING

A free-body diagram puts XJ on a dot and shows every force acting on it. For an object in space the list is short, which is what makes it predictable.

Read the physics: Free-body diagram

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Draw a dot or box for the object. Name each outside object that pushes or pulls it. Draw one arrow for each force on your chosen object.

02 HOW IT WORKS

Label each arrow with the source or force name. Longer arrows mean stronger forces when the scale is the same. Put velocity outside the force diagram.

03 WHAT TO CHECK

A hanging lamp has gravity from Earth and tension from the rope. Equal forces balance on the lamp. They are not a third-law pair: both act on the lamp.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 20 N lamp hangs at rest from one vertical rope. Draw and label its forces. Tension 20 N up equal-length arrows.

05 CONNECT TO XJ

A free-body diagram puts XJ on a dot and shows every force acting on it. For an object in space the list is short, which is what makes it predictable.

SAY IT BEFORE YOU WRITE

Which object receives every force in this diagram? I drew ___ because ___ acts on the chosen object.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Free-body diagram

ES: diagrama de cuerpo libre / PT: diagrama de corpo livre

Meaning in my words	How can you tell?
How would you explain free-body diagram to a friend? In my words, this means ___.	You are drawing a lamp hanging from a cable. What must be true of every arrow? I can tell because ___.
Example or sketch	**Non-example and why**
What arrows would you draw for a lamp hanging still? My example shows ___.	A student draws the lamp's downward pull on the cable on the lamp diagram. Where does it belong? I would change ___ because ___.

Selected object

ES: objeto elegido / PT: objeto escolhido

Meaning in my words	How can you tell?
How would you explain selected object to a friend? In my words, this means ___.	You circle only the cart. Which force belongs: the hand on the cart or the cart on the hand? I can tell because ___.
Example or sketch	**Non-example and why**
What would you put at the centre of a free-body diagram for a hanging lamp? My example shows ___.	A student switches to forces on the ceiling halfway through. How should they organize those? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Interacting object

ES: objeto que interactúa / PT: objeto que interage

Meaning in my words	How can you tell?
How would you explain interacting object to a friend? In my words, this means ___.	For a book on a table, what objects account for the upward and downward forces? I can tell because ___.
Example or sketch	**Non-example and why**
Which interacting object would you name beside the tension arrow on a hanging lamp? My example shows ___.	Someone labels a force arrow “the book’s speed.” What interacting object does that name? I would change ___ because ___.

Force source

ES: objeto que ejerce la fuerza / PT: objeto que exerce a força

Meaning in my words	How can you tell?
How would you explain force source to a friend? In my words, this means ___.	A floor pushes up on a crate. Which object is the source? I can tell because ___.
Example or sketch	**Non-example and why**
What full label would you write next to the weight arrow on the crate? My example shows ___.	A student says “right” is the source of a force. What does that word tell us instead? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a boundary around one object. Add force arrows with source labels; put motion information outside the diagram.

Labels you can use: object · boundary · outside.

Chosen object: _____

Tell a partner

I drew ___ because ___ acts on the chosen object.

Build the force overlay

15

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Keep only forces on the chosen object in its diagram.

CHOOSE ONE JOB TOGETHER

SYSTEMS ENGINEER ☐

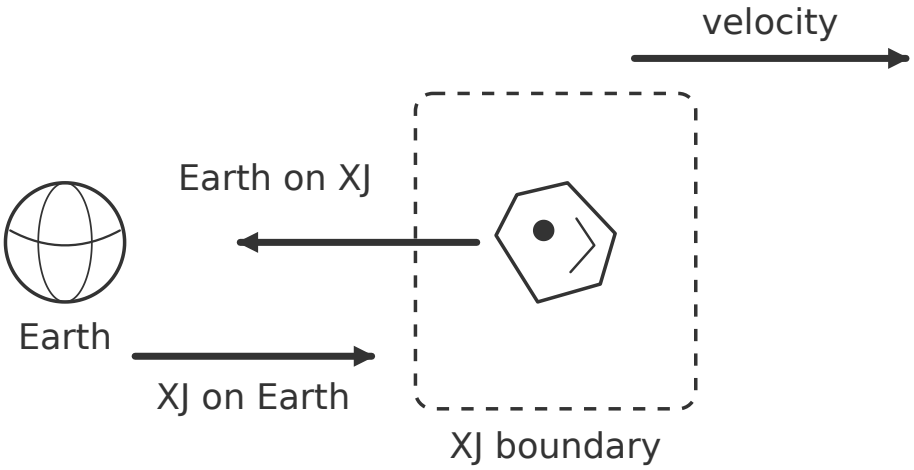
Outline XJ as the chosen object. Keep only the force arrows you would use to calculate XJ's acceleration.

ANIMATION DIRECTOR ☐

Separate the force and motion layers. Tell the animator which Earth arrow belongs in a different shot.

Word help: boundary / force / motion

The mission simulator needs one overlay called “forces on XJ.” Its proposed layers include Earth on XJ, XJ on Earth and XJ’s velocity.



01 OUR FIRST CALL

Which arrows belong in 'forces on XJ'? Explain what you keep and move.

This arrow acts on ___, so it belongs ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Where does the new arrow belong? Update the force layer and explain.

Now we would say ___ because ___.

Switch speakers. Does every arrow act on the object inside your boundary?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks